

mild 超関数のサーベイ

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1 はじめに

mild 超関数のモチベーションとかのサーベイ.

2 流体のやつ

$x \in \mathbf{R}^n$, $y \in \mathbf{R}$ とする. 超関数 $u(x, y)$ が $y > 0$ で定義されているときに, $y = 0$ での値, つまり境界値 $u(x, +0)$ は一般には定まらない. $u(x, +0)$ が $x_0 \in \mathbf{R}^n$ のある近傍 $N \subset \mathbf{R}^n$ で定義されるとき, $u(x, y)$ は $\{y = 0\}$ の正の側から $(x_0, 0)$ で mild であるという. このとき, $u(x, y)$ を $y \geq 0$ で定義された関数 $\text{ext}(u)(x, y)$ に拡張できる.

3 Yamazaki

We remark that in the case of a single differential equation, these results are given by Komatsu–Kawai [25] and Schapira [31]. Kataoka [23] extended to the case of microfunction solutions to a microdifferential equation by using the theory of mild microfunctions. Further, these facts can be extended to higher-codimensional boundary value problem for systems of differential equations (see for example Oaku–Yamazaki [28]). Moreover, elliptic and hyperbolic boundary value problems for systems are studied by Kashiwara–Kawai [12] and Kashiwara–Schapira [17] respectively from the microlocal point of view.

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